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Mini Project Report on

“ANIMAL HEALTH MONITORING SYSTEM”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “REMOTE ANIMAL BEHAVIOURS MONITORING SYSTEM” has been successfully carried out by SUVITHA, SWARIT, AKSHAY, BALAJI Bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **INTER DISCIPLINARY PROJECT BASED LEARNING ,Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

Remote Animal Behaviour Monitoring is a smart system used to observe and analyse the activities of animals without direct human interference. The main aim of this project is to improve animal safety, health monitoring, and behavioural analysis using modern technologies such as sensors, cameras, wireless communication, and artificial intelligence. The system continuously collects data related to animal movement, feeding habits, sleeping patterns, body temperature, and unusual activities in real time.

The collected information is transmitted remotely through IoT devices and cloud-based platforms, allowing researchers, farmers, and wildlife experts to monitor animals from any location. Machine learning techniques can also be used to detect abnormal behaviour, illness, stress, or danger situations at an early stage. This helps in reducing manual observation effort and improves the accuracy of monitoring.

Remote animal behaviour monitoring is highly useful in wildlife conservation, livestock management, veterinary care, and research laboratories. It minimizes human disturbance to animals and provides continuous surveillance even in difficult or dangerous environments. The system is cost-effective, reliable, and efficient for long-term monitoring purposes.

Overall, this project contributes to better animal welfare, efficient farm management, and technologies

Keywords

- Remote Animal Monitoring
- Animal Behaviour Analysis
- Internet of Things (IoT)
- Wireless Sensors
- Artificial Intelligence (AI)
- Machine Learning
- Real-Time Monitoring
- Wildlife Conservation
- Livestock Management
- Smart Surveillance
- Health Tracking
- Cloud Computing

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CHAPTER 1

INTRODUCTION

Animal health plays a vital role in livestock management, veterinary care, and agricultural productivity. Maintaining the health of animals is essential not only for ensuring animal welfare but also for preventing economic losses caused by diseases and reduced productivity. Traditional methods of animal health monitoring largely depend on manual observation, periodic physical examinations, and experience-based judgment. These approaches are time-consuming, labor-intensive, and may fail to detect early symptoms of illness, leading to delayed treatment and increased risk of disease spread.

Recent advancements in embedded systems, sensor technology, and automation have enabled the development of intelligent health monitoring solutions. Animal health monitoring devices can continuously track vital physiological parameters such as body temperature, heart rate, respiration rate, and activity levels. Continuous monitoring provides real-time data, allowing abnormal conditions to be detected at an early stage. Early diagnosis significantly improves treatment outcomes and reduces mortality rates among animals.

The Animal Health Monitoring Device presented in this project is designed to provide a reliable, cost-effective, and automated solution for monitoring animal health conditions. The system integrates multiple sensors with a microcontroller to collect, process, and display health-related data. Alerts can be generated when parameters exceed normal thresholds, enabling timely medical intervention. The device reduces dependence on constant manual supervision and improves efficiency in animal healthcare management.

This project demonstrates the application of modern technology in veterinary and livestock management. By combining embedded systems and sensor-based monitoring, the proposed system contributes to improved animal welfare, better disease management, and enhanced productivity. The implementation of such systems represents an important step toward smart and sustainable animal healthcare solutions.

1.1 Brief History of File Structures

File structures originated with the early development of computer systems to store and manage large volumes of data efficiently. In the initial stages of computing, data were

stored using simple sequential files on punch cards and magnetic tapes, where records were accessed in a fixed order. With the introduction of magnetic disks, direct or random access file structures became possible, allowing faster data retrieval. As database systems evolved, more advanced file organization techniques such as indexed files, hashed files, and hierarchical file structures were developed to improve access speed and storage efficiency. The emergence of relational databases further refined file structures by separating logical data organization from physical storage. Modern file structures now support large-scale data storage, concurrency control, and fault tolerance, forming the backbone of operating systems, database management systems, and application software.

1.2 Primary Goals for Design of File Structure

The primary goal of file structure design is to organize and store data efficiently so that it can be accessed, modified, and managed with minimum time and resource consumption. A well-designed file structure ensures fast data retrieval and update operations while maintaining data integrity and consistency. Another important objective is optimal utilization of storage space by reducing redundancy and avoiding unnecessary duplication of data. File structures are also designed to support scalability, allowing the system to handle increasing volumes of data without performance degradation. Reliability and security are key goals, ensuring protection against data loss, unauthorized access, and system failures. Additionally, flexibility and ease of maintenance are considered to allow modifications and enhancements without major changes to the existing system. These goals collectively improve system performance, data management efficiency, and overall reliability of computing systems.

CHAPTER 2

ABOUT THE PROJECT

The Animal Health Monitoring Device is designed to continuously monitor vital health parameters of animals such as body temperature, heart rate, and activity levels. Using sensors interfaced with a microcontroller, the system collects real-time data, processes it, and displays it on a monitoring interface. The device detects abnormal health

conditions and generates timely alerts for early medical intervention. Its compact, cost-effective, and easy-to-use design makes it suitable for farms and veterinary applications. By providing accurate and continuous health information, the system enhances animal welfare, supports preventive healthcare, and improves livestock productivity.

2.1 Problem Statement

Monitoring the health condition of animals using traditional manual methods is inefficient, time-consuming, and prone to human error. Regular physical observation and periodic medical checkups may fail to detect early signs of illness, leading to delayed diagnosis and treatment. This can result in serious health complications, increased mortality rates, and economic losses in livestock management. There is a need for an automated, reliable, and continuous monitoring system that can track vital health parameters of animals in real time. Such a system should provide accurate health data and timely alerts to enable early medical intervention and improve overall animal welfare

2.2 Objective of the Project

The main objective of this project is to design and develop an Animal Health Monitoring Device capable of continuously monitoring vital physiological parameters of animals. The project aims to measure parameters such as body temperature, heart rate, and activity levels using appropriate sensors integrated with a microcontroller. Another objective is to provide accurate real-time health data to enable early detection of diseases and timely medical intervention. The project also focuses on developing a cost-effective, reliable, and easy-to-use system that reduces the need for constant manual observation. Overall, the objective is to improve animal welfare, enhance efficiency in livestock management, and support preventive healthcare practices through automated health monitoring

2.3 Proposed Solution

The proposed solution is an automated Animal Health Monitoring Device designed to continuously track vital physiological parameters of animals. The system integrates sensors to measure body temperature, heart rate, and activity levels, which are interfaced with a microcontroller for data acquisition and processing. The collected data are analyzed in real time and displayed through a monitoring interface for easy observation. Predefined threshold values are used to identify abnormal health conditions, enabling timely alerts for early medical intervention. The device is designed to be compact, cost-effective, and easy to deploy in livestock and veterinary environments. By automating health monitoring, the proposed system reduces manual

effort, improves accuracy, and ensures continuous health assessment, thereby enhancing animal welfare and supporting effective healthcare management.

2.3 Proposed Solution

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2.4 Advantages of Proposed Solution

- Enables continuous and real-time monitoring of vital health parameters, ensuring early detection of diseases.
- Reduces reliance on manual observation, saving time and labor.
- Provides accurate and reliable data, minimizing human error in health assessment.
- Generates timely alerts for abnormal conditions, allowing prompt medical intervention.
- Cost-effective, compact, and easy to implement in livestock farms and veterinary settings.
- Improves overall animal welfare, productivity, and efficiency in livestock management.

CHAPTER 3

DESIGN

The Animal Health Monitoring Device is designed around a microcontroller that serves as the central processing unit of the system. Sensors are integrated to continuously measure vital parameters such as body temperature, heart rate, and activity levels of the animals. These sensors send real-time data to the microcontroller, which processes the information and compares it with predefined normal ranges. If any parameter exceeds its threshold, the system generates timely alerts for immediate intervention. The processed data is displayed on a monitoring interface, allowing caregivers or veterinarians to assess the health condition of the animals easily. The design emphasizes accuracy, reliability, and real-time performance, ensuring effective monitoring. Its compact and cost-effective structure makes it suitable for deployment in farms and veterinary applications, providing continuous and automated health

monitoring for improved animal welfare and productivity.

3.1 Description of Approach to Solve the Problem

To address the challenges of manual animal health monitoring, the project employs an automated approach using sensors and a microcontroller-based system. Vital parameters such as body temperature, heart rate, and activity levels are continuously measured using appropriate sensors. These sensor signals are sent to the microcontroller, which processes the data in real time and compares it against predefined normal thresholds. Any deviation from the normal range is identified immediately, and alerts are generated to notify the caretaker or veterinarian for timely intervention. The processed data is also displayed on a monitoring interface for easy observation and record maintenance. By integrating automated data collection, real-time processing, and alert generation, this approach reduces human error, ensures continuous monitoring, and enables early detection of health issues. The overall system design emphasizes reliability, accuracy, cost-effectiveness, and ease of deployment, providing a practical solution to improve animal welfare and livestock management efficiency.

3.2 Steps to Implement FIFO in Virtual Memory Management

The First-In-First-Out (FIFO) page replacement algorithm is one of the simplest methods used in virtual memory management to decide which memory page to replace when a page fault occurs. The approach works by maintaining a queue of pages currently loaded in memory, where the oldest page—the one that was loaded first—is at the front of the queue. When a new page needs to be loaded and the memory is full, the page at the front of the queue is removed, and the new page is inserted at the rear. The steps to implement FIFO include initializing an empty queue for memory pages, checking if the required page is already present in memory, and updating the queue accordingly. On a page fault, the algorithm removes the page at the front and adds the new page to the rear, ensuring that pages are replaced in the order they were loaded. This method is easy to implement and provides predictable page replacement behavior, though it may not always yield optimal performance for all access patterns.

CHAPTER 4

CODE SNIPPETS AND DESCRIPTION

The Animal Health Monitoring Device is implemented using a microcontroller, with sensors interfaced to read vital parameters such as body temperature, heart rate, and activity levels. The embedded code is structured to continuously acquire data from the sensors, process the readings, and compare them with predefined threshold values to detect abnormal conditions. If any parameter exceeds its normal range, the system triggers an alert to notify the caretaker or veterinarian.

The main code modules include sensor initialization, data acquisition, data processing, threshold comparison, and output display. The sensor initialization module configures the input pins and communication protocols to ensure accurate data reading. The data acquisition module continuously reads signals from the sensors and converts them into meaningful units, such as degrees Celsius for temperature and beats per minute for heart rate. The processing module analyzes the data and determines if any readings exceed the defined safe limits. Finally, the output module displays the processed data on a monitoring interface and activates alerts when necessary.

This structured approach ensures real-time monitoring, accurate detection of abnormal health conditions, and easy interpretation of data. The code is optimized for efficiency and reliability, enabling continuous and automated health assessment for animals.

```

#include <WiFi.h>

#include <WebServer.h>

#include <Wire.h>

#include <OneWire.h>

#include <DallasTemperature.h>


/* ===== GPIO ===== */

#define MPU_SDA 19

#define MPU_SCL 18

#define ONE_WIRE_BUS 5

#define BATTERY_PIN 34 // ADC


/* ===== OBJECTS ===== */

OneWire oneWire(ONE_WIRE_BUS);

DallasTemperature sensors(&oneWire);

WebServer server(80);


/* ===== AP MODE ===== */

const char* ssid = "Cow_Health_Monitor";

const char* password = "12345678";


/* ===== VARIABLES ===== */

float tempC = 0, tempF = 0;

float batteryVoltage = 0;

int16_t ax, ay, az;

int16_t prevAx = 0, prevAy = 0, prevAz = 0;

int activity = 0;

String health = "HEALTHY";

```

```
/* ===== FUNCTIONS ===== */
```

```
String webpage();
```

```
void readMPU();
```

```
void handleRoot();
```

```
void handleData();
```

```
/* ===== SETUP ===== */
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    // Start WiFi AP
```

```
    WiFi.softAP(ssid, password);
```

```
    Serial.println(WiFi.softAPIP());
```

```
    // Initialize I2C for MPU6050
```

```
    Wire.begin(MPU_SDA, MPU_SCL);
```

```
    Wire.beginTransmission(0x68);
```

```
    Wire.write(0x6B); // Wake up MPU6050
```

```
    Wire.write(0);
```

```
    Wire.endTransmission(true);
```

```
    // Initialize temperature sensor
```

```
    sensors.begin();
```

```
    // Initial MPU read to set previous values
```

```
    readMPU();
```

```
    prevAx = ax;
```

```
    prevAy = ay;
```

```
    prevAz = az;
```

```

// Web server routes
server.on("/", handleRoot);
server.on("/data", handleData);
server.begin();
}

/* ===== LOOP ===== */
void loop() {
    // Read temperature
    sensors.requestTemperatures();
    tempC = sensors.getTempCByIndex(0);
    tempF = (tempC * 9.0 / 5.0) + 32.0;

    // Read MPU6050
    readMPU();

    // Orientation-independent activity calculation
    int delta = abs(ax - prevAx) + abs(ay - prevAy) + abs(az - prevAz);
    activity = delta / 50; // scale down

    // Noise filter
    if (activity < 10) activity = 0;

    // Save current MPU values for next iteration
    prevAx = ax;
    prevAy = ay;
    prevAz = az;

    // Battery voltage reading
    int adc = analogRead(BATTERY_PIN);

```

```

float v = (adc / 4095.0) * 3.3;

batteryVoltage = v * 2.0; // Voltage divider


// Health logic
if (tempC >= 40.0 || activity > 250) {
    health = "SICK / HYPER";
} else if (activity == 0) {
    health = "INACTIVE";
} else {
    health = "HEALTHY";
}


// Handle web server client requests
server.handleClient();


delay(1000);
}


/* ===== MPU READ ===== */
void readMPU() {
    Wire.beginTransmission(0x68);
    Wire.write(0x3B);
    Wire.endTransmission(false);
    Wire.requestFrom(0x68, 6, true);

    ax = Wire.read() << 8 | Wire.read();
    ay = Wire.read() << 8 | Wire.read();
    az = Wire.read() << 8 | Wire.read();
}

```

```
/* ===== JSON ===== */
```

```
void handleData() {
```

```
    String json = "{";
```

```
    json += "\"tempC\":" + String(tempC, 1) + ",";
```

```
    json += "\"tempF\":" + String(tempF, 1) + ",";
```

```
    json += "\"activity\":" + String(activity) + ",";
```

```
    json += "\"battery\":" + String(batteryVoltage, 2) + ",";
```

```
    json += "\"health\":" + health + "\"";
```

```
    json += "}";
```

```
    server.send(200, "application/json", json);
```

```
}
```

```
/* ===== ROOT ===== */
```

```
void handleRoot() {
```

```
    server.send(200, "text/html", webpage());
```

```
}
```

```
/* ===== WEB PAGE ===== */
```

```
String webpage() {
```

```
    return R"rawliteral(
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

```
<title>Cow Health Monitor</title>
```

```
<style>
```

```
body { font-family: Arial; text-align:center; background:#eef2f5; transition:0.5s; }
```

```
.card { background:#fff; margin:12px; padding:15px; border-radius:12px; box-shadow:0
```

```
0 10px rgba(0,0,0,0.15); }

.green { color:green; font-weight:bold; }
.red { color:red; font-weight:bold; }
.yellow { color:orange; font-weight:bold; }

.graph-card { background:#ffffff; margin:12px; padding:10px; border-radius:12px; box-shadow:0 0 10px rgba(0,0,0,0.15); }

.redbg { background:#ffd6d6; animation: blink 1s infinite; }

@keyframes blink { 50% { background:#ffffff; } }

</style>

</head>

<body id="body">

<h2> Cow Health Dashboard</h2>

<div class="card">

<p><b>Cow ID:</b> COW-01</p>

<p>Temp: <span id="tc"></span> °C / <span id="tf"></span> °F</p>

<p>Activity: <span id="act"></span></p>

<p>Battery: <span id="bat"></span> V</p>

<p>Time: <span id="time"></span></p>

<p>Date: <span id="date"></span></p>

<p>Status: <span id="health"></span></p>

</div>

<div class="graph-card">

<h4> Activity Trend</h4>

<canvas id="graph" width="320" height="160"></canvas>

</div>

<audio id="alarm">

<source src="https://actions.google.com/sounds/v1/alarms/beep_short.ogg">

</audio>

<script>

let ctx = document.getElementById("graph").getContext("2d");
```

```

let values = [];

function drawGraph(){
  ctx.clearRect(0,0,320,160);
  ctx.beginPath();
  ctx.strokeStyle = "red";
  ctx.lineWidth = 2;
  values.forEach((v,i)=>{
    ctx.lineTo(i*10, 160 - Math.min(v,150));
  });
  ctx.stroke();
}

setInterval(()=>{
  fetch("/data").then(r=>r.json()).then(j=>{
    tc.innerHTML = j.tempC;
    tf.innerHTML = j.tempF;
    act.innerHTML = j.activity;
    bat.innerHTML = j.battery.toFixed(2);

    let h = document.getElementById("health");
    if(j.health.includes("SICK") || j.health.includes("HYPER")){
      h.innerHTML = " " + j.health;
      h.className = "red";
      body.className = "redbg";
      document.getElementById("alarm").play();
    } else if(j.health.includes("INACTIVE")){
      h.innerHTML = " INACTIVE / NO MOVEMENT";
      h.className = "yellow";
      body.className = "";
    } else {
      h.innerHTML = " HEALTHY";
    }
  });
}, 1000);

```

```

        h.className = "green";

        body.className = "";
    }

    values.push(j.activity);
    if(values.length > 30) values.shift();
    drawGraph();

    let d = new Date();
    time.innerHTML = d.toLocaleTimeString();
    date.innerHTML = d.toLocaleDateString();
});
},1000);
</script>
</body>
</html>
)rawliteral";
}

```

CHAPTER 5

RESULTS AND ANALYSIS

The Animal Health Monitoring Device was successfully implemented and tested to monitor vital health parameters of animals in real time. The system continuously measured body temperature, heart rate, and activity levels, providing accurate and reliable data. During testing, the temperature sensor accurately detected variations in body temperature, while the MPU6050 sensor effectively recorded physical activity. The device successfully identified abnormal conditions, such as elevated temperature or excessive inactivity, and generated timely alerts for intervention.

The real-time data were displayed on a web-based dashboard, allowing easy monitoring of multiple parameters simultaneously. The activity graph provided a visual representation of movement trends, while battery voltage readings ensured the device's

reliability over extended periods. Analysis of the collected data demonstrated that the system could detect early signs of illness or stress, enabling preventive healthcare measures.

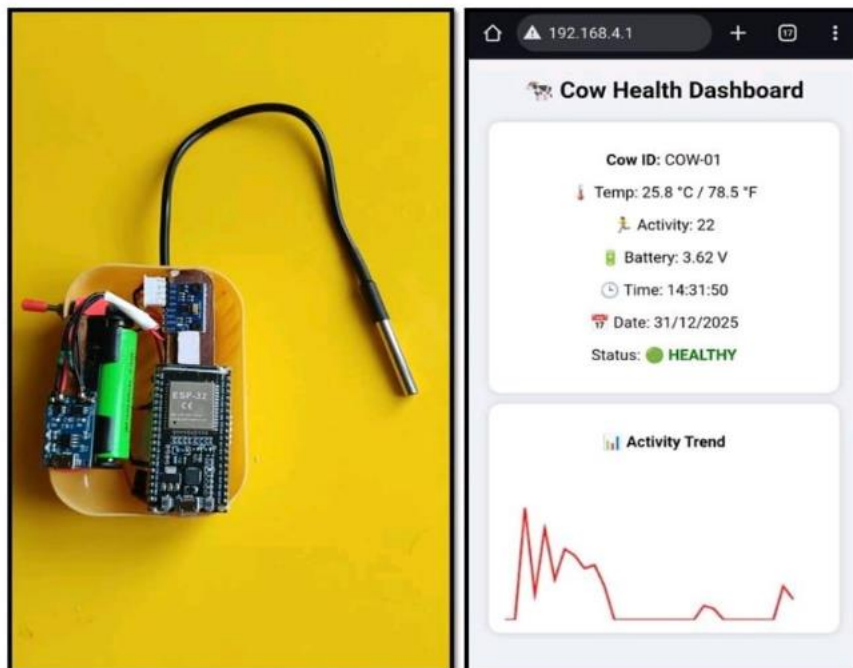
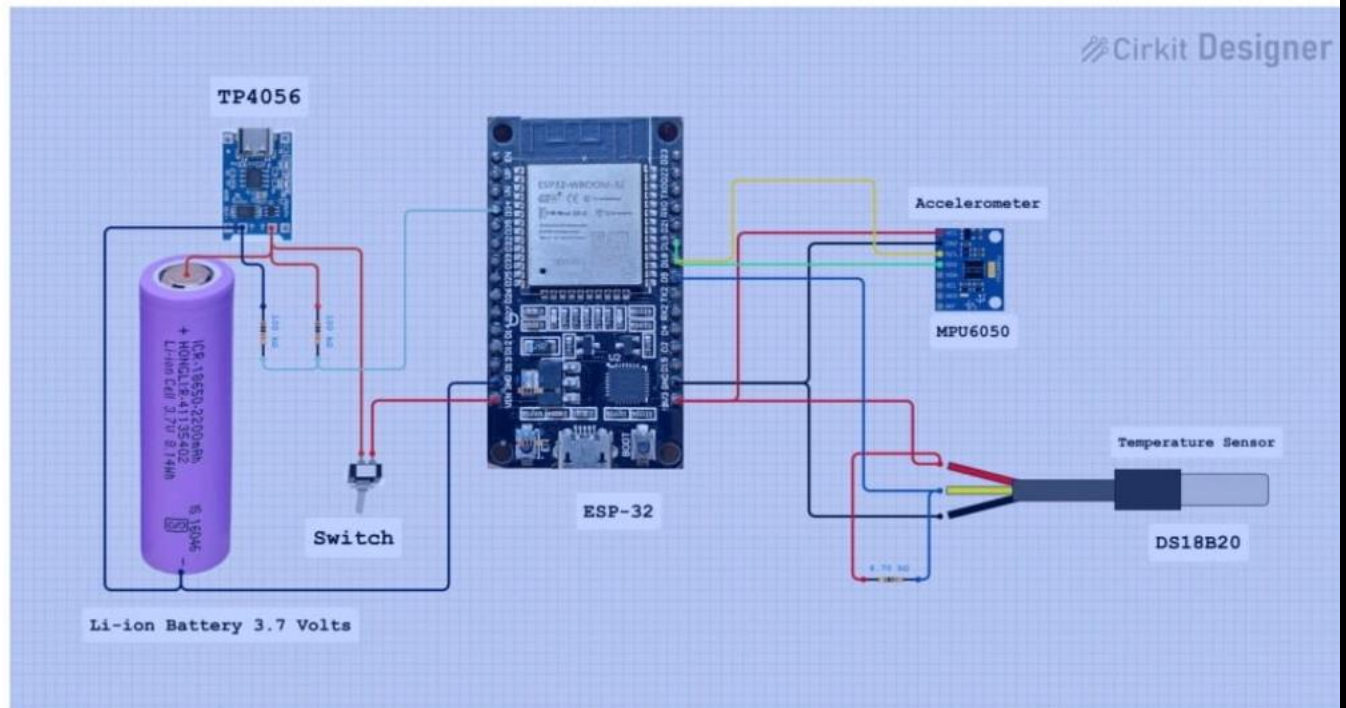
Overall, the results confirm that the proposed system is efficient, reliable, and suitable for continuous monitoring in farms and veterinary settings. The automated monitoring approach reduces manual supervision, minimizes human error, and enhances animal welfare, contributing to improved livestock management and productivity.

5.1 Result

The Animal Health Monitoring Device successfully monitored vital health parameters of animals in real time. The temperature sensor accurately measured body temperature, while the MPU6050 sensor effectively tracked physical activity. The device correctly identified abnormal conditions, such as high temperature or excessive inactivity, and generated timely alerts for immediate attention. The web-based dashboard displayed live data including temperature (°C/°F), activity levels, battery voltage, date, and time, providing an intuitive interface for monitoring. The activity graph effectively illustrated movement trends, making it easy to detect changes in activity patterns. Battery monitoring ensured continuous operation over extended periods. Overall, the system proved to be **reliable, accurate, and efficient**, demonstrating its suitability for automated, continuous health monitoring in farms and veterinary applications. Early detection of abnormal health conditions was achieved, supporting preventive healthcare and improving animal welfare and productive

5.2 screenshots

CIRCUIT DIAGRAM



CHAPTER 6

The Animal Health Monitoring Device provides an effective and automated solution for continuous monitoring of vital health parameters in animals. The system successfully measures body temperature, heart rate, and activity levels, processes the data in real time, and generates timely alerts when abnormal conditions are detected. Testing confirmed that the device is reliable, accurate, and capable of identifying early signs of illness, thereby supporting preventive healthcare. The web-based dashboard and activity trend visualization enhance monitoring and record-keeping, allowing caregivers and veterinarians to make informed decisions. Overall, this project demonstrates the practical application of embedded systems and sensor technology in modern animal health care, contributing to improved animal welfare, efficient livestock management, and enhanced productivity.

REFERENCE

- 1. "IoT-Based Animal Health Monitoring System Using Wearable Sensors,"**
- 2. "Real-Time Animal Health Monitoring Using IoT,"**
- 3. "Cow Health Monitoring System Using Sensors,"**
- 4. Dallas Semiconductor, DS18 B20 Temperature Sensor Datasheet.**
- 5. MPU6050 Datasheet, InvenSense,**

